

Character codegrees of VZ and Camina p -groups

Ayush Udeep

ABSTRACT. The character codegree of an irreducible character of a finite group G is given by the index of its kernel in G upon the character degree. We compute the character codegrees for VZ and Camina p -groups, and also obtain the character codegrees set of p -groups of order $\leq p^5$, where p is an odd prime.

1. Introduction

In this article, G denotes a finite p -group, where p denotes a prime. The character codegree of an irreducible character χ of G is the index of the kernel of χ in G upon the degree of χ . We denote the character codegree of χ by $\text{cod}(\chi)$, and the set of character codegrees of all irreducible characters of G , all linear characters of G and all nonlinear irreducible characters of G by $\text{cod}(G)$, $\text{cod}_{\text{lin}}(G)$ and $\text{cod}_{\text{nl}}(G)$, respectively. Qian et al. [22] initiated the study of the irreducible character codegrees of G . In recent past, various researchers have obtained interesting structural results for various classes of non-abelian p -groups by studying their character codegree sets. As it turns out, certain versions of character codegrees of known results for character degrees hold for some classes of p -groups. For example, Du and Lewis [7] proved that the nilpotence class of G is bounded in terms of the largest codegree for an irreducible character of G . They also proved that a p -group with exactly two codegrees is elementary abelian, and a p -group with exactly three codegrees has nilpotence class at most 2. In [5], Croome and Lewis studied p -groups with exactly four codegrees. In [4], Croome and Lewis proved that if a non-abelian group G has order p^n and $\text{cod}(G)$ contains every power of p up to p^{n-1} , then the nilpotence class of G can either be 2 or $n - 1$. Moretó [16] proved that the nilpotence class of a p -group is bounded in terms of the number of character degrees and the number of character codegrees.

On the other hand, there are certain properties of character degrees that character codegrees do not respect. For example, while isoclinism preserves the set of irreducible character degrees of G , denoted $\text{cd}(G)$, it does not preserve $\text{cod}(G)$. For example, consider groups $\phi_2(211)a$ and $\phi_2(211)b$ of order p^4 ($p > 3$) belonging to the isoclinic family Φ_2 (see [9] for the classification of p -groups of order p^4 ($p > 3$)). We have $\text{cod}(\phi_2(211)a) = \{1, p, p^2\}$ whereas $\text{cod}(\phi_2(211)b) = \{1, p, p^3\}$ (see Table 1). Recently, in [20], Prajapati and Udeep obtained a relation between the minimal faithful permutation representation degree of a p -group of odd order and its character codegrees. We wish to explore other such properties of the character codegrees for various classes of finite p -groups.

In this article, we obtain the character codegrees sets for certain interesting classes of finite p -groups. In Section 2, we compute the character codegrees sets for abelian, VZ and Camina p -groups. For an abelian p -group A , we obtain $\text{cod}(A) = \{p^i : i = 0, 1, \dots, \log_p(\exp(A))\}$ (see Corollary 2.2). A group is called a VZ-group if all its nonlinear irreducible complex characters vanish off the center. VZ-groups were studied in [8, 11, 12]. We obtain an algorithm for the computation of $\text{cod}(G)$ when G is any general VZ-group (see Remark 2.3). We observe that the problem of computation of $\text{cod}(G)$, in this case, reduces to finding

2020 *Mathematics Subject Classification.* Primary 20D15; secondary 20C15, 20B05.

Key words and phrases. character codegree, VZ p -groups, groups of order less than p^6 .

the exponent of G/G' and obtaining the set

$$\left\{ |N| : N \leq Z(G) \text{ such that } G' \not\leq N \text{ and } \frac{Z(G)}{N} \text{ is cyclic} \right\}.$$

Next, we compute the exact character codegrees for certain sub-classes of VZ p -groups. In fact, we prove the following.

THEOREM 1.1. *Let G be a VZ p -group.*

- (i) *If $Z(G)$ is cyclic, then $\text{cod}(G) = \{p^i, (|G|Z(G))^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.*
- (ii) *If $Z(G)$ is elementary abelian, then $\text{cod}(G) = \{p^i, p \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.*
- (iii) *If $d(G') = d(Z(G))$, then*

$$\text{cod}(G) = \{p^i, p^{l_j} \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G')), j = 1, 2, \dots, r\},$$

where $Z(G) \cong C_{p^{l_1}} \times C_{p^{l_2}} \times \dots \times C_{p^{l_r}}$ ($l_1 \geq l_2 \geq \dots \geq l_r$).

- (iv) *If $d(G') < d(Z(G)) = 2$, then either*

$$\text{cod}(G) = \{p^i, p^j \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G')), j = l_2, l_2 + 1, \dots, l_1\},$$

or

$$\text{cod}(G) = \{p^i, \exp(Z(G)) \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$$

where $Z(G) \cong C_{p^{l_1}} \times C_{p^{l_2}}$ ($l_1 \geq l_2$).

As a corollary of Theorem 1.1, we prove that for an extraspecial p -group G , $\text{cod}(G) = \{1, p, (p \cdot |G|)^{1/2}\}$ (see Corollary 2.4). We also obtain the character codegrees set for Camina p -groups. A p -group G is called a Camina p -group if the conjugacy class of g is gG' for all $g \in G \setminus G'$ [6, 10]. Dark and Scoppola [6] proved that Camina p -groups are nilpotent of class at most 3. For a Camina p -group G , we prove that if the nilpotency class of G is 2, then $\text{cod}(G) = \{1, p, p \cdot |G/Z(G)|^{1/2}\}$, otherwise, $\text{cod}(G) = \{1, p, p \cdot |G/G'|^{1/2}, p \cdot |G/Z(G)|^{1/2}\}$ (see Theorem 2.5).

In Section 3, we obtain the character codegrees set for all p -groups of order $\leq p^5$. Groups of order $\leq p^5$, where $p \geq 3$, are classified in certain isoclinic families, and listed in [9]. Since isoclinism does not preserve $\text{cod}(G)$, the task of computing the character codegrees set of p -groups collectively becomes a little difficult. We obtain general results for groups of order p^5 ($p > 3$) in Lemma 3.2 and 3.3. By utilizing them along with Theorem 1.1 and Corollary 2.4, we compute the character codegrees set for all p -groups of order less than p^6 ($p > 3$) in Section 3, and list them in Table 2 and 3. We also construct MAGMA [3] code `CharacterCodegree` to obtain $\text{cod}(G)$ for a given group G . We utilize it to verify the data of Tables 1, 2 and 3 for small primes ≥ 5 , and to compute $\text{cod}(G)$ when G is a non-abelian group of order 3^n for $n = 3, 4, 5$ on MAGMA. We summarize the results of Section 3 in Theorem 3.12.

Before we start, here we mention a list of notations (other than those already mentioned above) that we use throughout the article. For a finite group G ,

G'	the commutator subgroup of G
$\text{cl}(G)$	the nilpotency class of G
$\exp(G)$	the exponent of G
$\text{Core}_G(H)$	the core of H in G , for $H \leq G$
$\text{Irr}(G)$	the set of irreducible complex characters of G
$\text{lin}(G)$	the set of linear characters of G
$\text{nl}(G)$	the set of non-linear irreducible characters of G
$\text{cd}(G)$	the character degree set of G , i.e., $\{\chi(1) \mid \chi \in \text{Irr}(G)\}$
$\text{Irr}(G N)$	$\text{Irr}(G) \setminus \text{Irr}(G/N)$, where N is a normal subgroup of G

2. Character Codegrees Sets of VZ p -groups and Camina p -groups

We begin with the following lemma.

LEMMA 2.1. *Let A be an abelian p -group of exponent p^a and M be a normal subgroup of A . If A/M is cyclic then $|M| \geq |A|/p^a$. Further, for each b ($0 \leq b \leq a$), there exists a normal subgroup N of A such that $A/N \cong C_{p^b}$.*

PROOF. First, note that if $|M| < |A|/p^a$, then $|A/M| > p^a$. Hence, A/M cannot be cyclic as it has no element of order greater than p^a . Thus, if A/M is cyclic then $|M| \geq |A|/p^a$.

To prove the second part of the lemma, suppose

$$A = \langle a_1, a_2, \dots, a_s \rangle \cong C_{p^a} \times C_{p^{r_2}} \times \dots \times C_{p^{r_s}},$$

where r_i 's are positive integers such that $a = r_1 \geq r_2 \geq \dots \geq r_s$. To find a normal subgroup N of A of order $|A|/p^b$, for each $0 \leq b \leq a$, so that A/N is cyclic, we use the following algorithm:

- (1) Let $X = [a = r_1, r_2, \dots, r_s]$ be an ordered set, i.e., $X[j] = r_j$ for some $1 \leq j \leq s$.
- (2) Take $i = s$.
- (3) If $X[i] \geq b$, then

$$N = \langle a_1, a_2, \dots, a_{s-1}, a_s^{p^{r_s-b}} \rangle.$$

- (4) Otherwise, if $X[s] \geq b$, then $i = i - 1$.
- (5) Continue till $i = 1$.

Since $b \leq a$, existence of such an N is guaranteed after some steps of the algorithm. $\square$

We derive the following corollary with the help of Lemma 2.1.

COROLLARY 2.2. *Let A be an abelian p -group. Then $\text{cod}(A) = \{p^i \mid i = 0, 1, \dots, \log_p(\exp(A))\}$.*

PROOF. Since A is abelian, all the irreducible characters are linear. Then $A/\ker(\chi)$ is cyclic, for each $\chi \in \text{Irr}(A)$. Hence we consider those normal subgroups N of A such that A/N is cyclic. By Lemma 2.1, $|\ker(\chi)| \geq |A|/p^a$ so that $\text{cod}(\chi) \leq p^a$, for each χ . Further, for each b ($0 \leq b \leq a$), there exists $\chi \in \text{Irr}(A)$ such that $|\ker(\chi)| = |A|/p^b$. That is, for each b ($0 \leq b \leq a$), there exists $\chi \in \text{Irr}(A)$ so that $\text{cod}(\chi) = |A/\ker(\chi)| = p^b$. Therefore, $\text{cod}(A) = \{p^b \mid 0 \leq b \leq a\}$. $\square$

A group G is called a VZ p -group if all its nonlinear irreducible complex characters vanish off the center. Hence, $\text{nl}(G) = \{|G/Z(G)|^{1/2}\lambda \mid \lambda \in \text{Irr}(Z(G)) \text{ such that } G' \not\leq \ker(\lambda)\}$. A VZ p -group is nilpotent of class 2, and hence, the commutator subgroup of the group is contained in the center (see [12]). In fact, G is a VZ-group, if and only if $\text{cd}(G) = \{1, |G/Z(G)|^{1/2}\}$ (see [8, Theorem A]). Further, from [13, Lemma 2.4], $G/Z(G)$ and G' are elementary abelian p -groups. In Remark 2.3, we obtain an algorithm for the computation of $\text{cod}(G)$ when G is a VZ p -group.

REMARK 2.3. *(Algorithm for the computation of $\text{cod}(G)$ when G is a VZ p -group)*

Let G be a VZ p -group. By Corollary 2.2, $\text{cod}_{\text{lin}}(G) = \{p^i : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$. Thus, $\text{cod}_{\text{lin}}(G)$ depends on the exponent of G/G' . Since $\chi(1) = |G/Z(G)|^{1/2}$ for all $\chi \in \text{nl}(G)$, computation of $\text{cod}_{\text{nl}}(G)$ depends solely on the possibilities of the order of $\ker(\chi)$. That is, for each $\chi \in \text{nl}(G)$,

$$\text{cod}(\chi) = \frac{|G/\ker(\chi)|}{\chi(1)} = \frac{|G/Z(G)||Z(G)/\ker(\chi)|}{|G/Z(G)|^{1/2}} = |G/Z(G)|^{1/2} \left| \frac{Z(G)}{\ker(\chi)} \right|.$$

Clearly, $\ker(\chi) \leq Z(G)$ for all $\chi \in \text{nl}(G)$. Thus,

$$\text{cod}_{\text{nl}}(G) = \left\{ |G/Z(G)|^{1/2} \left| \frac{Z(G)}{N} \right| : N \leq Z(G) \text{ such that } G' \not\leq N \text{ and } \frac{Z(G)}{N} \text{ is cyclic} \right\}.$$

Finally, $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \text{cod}_{\text{nl}}(G)$.

By Remark 2.3, it is clear that we may need more information about the group structure to compute the exact $\text{cod}(G)$ for a general VZ-group. Now, we prove Theorem 1.1, where we utilize the properties of some specific classes of VZ p -groups to obtain their respective character codegrees sets.

PROOF OF THEOREM 1.1. Let G be a VZ p -group. By Corollary 2.2, $\text{cod}_{\text{lin}}(G) = \{p^i : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$. We now investigate the set $\text{cod}_{\text{nl}}(G)$ for the cases (i), (ii) and (iii). For that, we

have to find out the orders of those subgroups N of $Z(G)$ which do not contain G' , and that their quotient in $Z(G)$ is cyclic.

First, suppose that $Z(G)$ is cyclic. Since G' is elementary abelian and is contained in $Z(G)$, we must have $G' \cong C_p$. If N is a nontrivial subgroup of $Z(G)$, then N must contain G' . Such an N is not a suitable candidate for $\ker(\chi)$. Hence, N must be the trivial subgroup of $Z(G)$, that is, $\ker(\chi) = 1$ for all $\chi \in \text{nl}(G)$. Thus, $\text{cod}(\chi) = |G/\ker(\chi)|/\chi(1) = |G|/|G/Z(G)|^{1/2} = (|G||Z(G)|)^{1/2}$, for all $\chi \in \text{nl}(G)$. Finally, $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \text{cod}_{\text{nl}}(G) = \{p^i, (|G||Z(G)|)^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.

Next, suppose that $Z(G)$ is elementary abelian, of rank r . Now, the required subgroup N of $Z(G)$ must satisfy $Z(G)/N \cong C_p$ which implies that $|N| = |Z(G)|/p = p^{r-1}$. Thus, for N , choose a subgroup of $Z(G)$ of rank $r-1$ that does not contain G' . Note that there are no other possibilities for the order of such a subgroup N . Then, for all $\chi \in \text{nl}(G)$, $|\ker(\chi)| = |N| = |Z(G)|/p$, and thus, $\text{cod}(\chi) = p \cdot |G/Z(G)|^{1/2}$. Finally, $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \text{cod}_{\text{nl}}(G) = \{p^i, p \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.

Next, suppose that $d(G') = d(Z(G))$ and $Z(G) = \langle a_1, a_2, \dots, a_r \rangle \cong C_{p^{l_1}} \times C_{p^{l_2}} \times \dots \times C_{p^{l_r}}$ where $l_1 \geq l_2 \geq \dots \geq l_r$. Now, for the required subgroup N , $Z(G)/N$ must be cyclic, thus $|Z(G)/N| \leq p^{l_1}$ which implies that $|N| \geq p^{l_2 + \dots + l_r}$. Further $d(N) < r$ since G' must not be contained in N . Hence, $d(N)$ must be $r-1$. This forces that

$$|N| = p^{\sum_{k=1, k \neq j}^r l_k}, \text{ for some } 1 \leq j \leq r.$$

Conversely, if, for each $1 \leq j \leq r$, we take $N = \langle a_1, \dots, a_{j-1}, a_{j+1}, \dots, a_r \rangle$, then $|N| = p^{\sum_{k=1, k \neq j}^r l_k}$, for each such j . Thus, if $\chi \in \text{nl}(G)$, then $|\ker(\chi)|$ belongs to the set

$$\{p^{\sum_{k=1, k \neq j}^r l_k} : j = 1, 2, \dots, r\}.$$

Hence, $\text{cod}(\chi) = |G/\ker(\chi)|/\chi(1) = \frac{|G/Z(G)||Z(G)/\ker(\chi)|}{|G/Z(G)|^{1/2}}$. Thus,

$$\text{cod}_{\text{nl}}(G) = \{p^{l_j} \cdot |G/Z(G)|^{1/2} : j = 1, 2, \dots, r\}.$$

Finally, we get $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \text{cod}_{\text{nl}}(G) = \{p^i, p^{l_j} \cdot |G/Z(G)|^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G')), j = 1, 2, \dots, r\}$.

Finally, suppose $d(G') < d(Z(G)) = 2$. Since G is a VZ p -group, G' must be cyclic of order p . We have two cases here.

Case I: We can choose generators of $Z(G)$ such that $Z(G) = \langle \alpha, \beta \rangle \cong C_{p^{l_1}} \times C_{p^{l_2}}$ ($l_1 \geq l_2$) and $G' = \langle \beta^{p^{l_2-1}} \rangle \cong C_p$. Now, for the required subgroup N , $Z(G)/N$ must be cyclic, thus $|Z(G)/N| \leq p^{l_1}$ which implies that $|N| \geq p^{l_2}$. For each k , where $l_2 \leq k \leq l_1$, consider the subgroups $N_k = \langle \alpha^{p^{l_1-k}} \beta \rangle \cong C_{p^k}$. Then for each $l_2 \leq k \leq l_1$, G' is not contained in N_k , and $Z(G)/N_k \cong C_{p^{l_1+l_2-k}}$. Thus, order of the kernel of a nonlinear irreducible character χ has the following possibilities:

$$p^k \text{ for } k = l_2, l_2 + 1, \dots, l_1.$$

Then $\text{cod}(\chi) = |G/Z(G)|^{1/2} \cdot |Z(G)/N|$ has the possibilities $|G/Z(G)|^{1/2} \cdot p^{l_1+l_2-k}$ for $k = l_2, l_2 + 1, \dots, l_1$. Thus,

$$\text{cod}_{\text{nl}}(G) = \{p^j \cdot |G/Z(G)|^{1/2} : j = l_2, l_2 + 1, \dots, l_1\}.$$

Case II: We can choose generators of $Z(G)$ such that $Z(G) = \langle \alpha, \beta \rangle \cong C_{p^{l_1}} \times C_{p^{l_2}}$ ($l_1 \geq l_2$) and $G' = \langle \alpha^{p^{l_1-1}} \rangle \cong C_p$. Again, for the required subgroup N (here $d(Z(G)) = 2$), N and $Z(G)/N$ must be cyclic, thus $|Z(G)/N| \leq p^{l_1}$ which implies that $|N| \geq p^{l_2}$. Let g be an element of $Z(G)$ of order greater than p^{l_2} . Then $g = \alpha^i \beta^j$ where i is chosen so that the order of α^i is greater than p^{l_2} . Then $\alpha^{p^{l_1-1}} \in \langle \alpha^i \beta^j \rangle$, and hence, G' is contained in any cyclic subgroup of order greater than p^{l_2} . The only possibility for the order of the required subgroup N is p^{l_2} . Consider $N = \langle \beta \rangle \cong C_{p^{l_2}}$. Then N and $Z(G)/N$ are cyclic, and G' is not contained in N . Hence, $|\ker(\chi)|$ must be p^{l_2} for all $\chi \in \text{nl}(G)$. Hence, $\text{cod}(\chi) = |G/Z(G)|^{1/2} \cdot p^{l_1+l_2-l_2} = p^{l_1} \cdot |G/Z(G)|^{1/2} = \exp(Z(G)) \cdot |G/Z(G)|^{1/2}$, for all $\chi \in \text{nl}(G)$. Thus, in this case, we have

$$\text{cod}_{\text{nl}}(G) = \{\exp(Z(G)) \cdot |G/Z(G)|^{1/2}\}.$$

Since $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \text{cod}_{\text{nl}}(G)$ where $\text{cod}_{\text{lin}}(G) = \{p^i : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$, we obtain the desired result. This concludes the proof. $\square$

With the help of Theorem 1.1, we obtain $\text{cod}(G)$ when G is an extraspecial p -group. Note that a p -group G is called extraspecial if $Z(G) = G'$ and the quotient $G/Z(G)$ is a non-trivial elementary abelian p -group. It is easy to see that every extraspecial p -group is a VZ p -group. We have the following corollary.

COROLLARY 2.4. *Let G be an extraspecial p -group. Then $\text{cod}(G) = \{1, p, (p \cdot |G|)^{1/2}\}$.*

PROOF. Since G is VZ as well, $Z(G) = G'$ is cyclic of order p and G/G' is elementary abelian. By Theorem 1.1, $\text{cod}(G) = \{p^i, (|G||Z(G)|)^{1/2} : i = 0, 1, \dots, \log_p(\exp(G/G'))\} = \{1, p, (p \cdot |G|)^{1/2}\}$. $\square$

Now, we deal with the Camina p -groups. First, we mention the preliminaries we utilize to obtain the character codegrees set. The pair (G, N) is said to be a Camina pair if for every $g \in G \setminus N$, $gN \subseteq \text{Cl}_G(g)$, the conjugacy class of g . It is easy to see that $Z(G) \leq N \leq G'$. When $N = G'$, the group G is called a Camina group. By [15, Lemma 3], if G is a Camina group, then for all $g \in G \setminus G'$, $\chi(g) = 0$ for all $\chi \in \text{nl}(G)$. Dark and Scoppola [6] proved that nilpotency class of G is at most 3. If G is a Camina p -group of class 2, then $G' \leq Z(G)$. Thus, $G' = Z(G)$, and hence by [15, Lemma 3] and [14, Corollary 2.3], G is a VZ p -group such that $Z(G)$ and G/G' are elementary abelian. If G is a Camina p -group of class 3, then $(G, Z(G))$ is a Camina pair, and $|G/G'| = p^{2n}$, $|G'/Z(G)| = p^n$ and $|G/Z(G)| = p^{3n}$, where n is even (see [14, Theorem 5.2]). Further, by [14, Corollary 2.3 and 5.3], G/G' , $G'/Z(G)$ and $Z(G)$ are elementary abelian p -groups. Moreover, $\text{nl}(G) = \text{Irr}(G|Z(G)) \sqcup \text{nl}(G/Z(G))$ as a disjoint union and $\text{cd}(G) = \{1, p^n, p^{3n/2}\}$ (see [18, Section 3, pp. 51–52] for the detailed discussion). We obtain the following result for Camina p -groups.

THEOREM 2.5. *Let G be a Camina p -group.*

- (1) *If the nilpotency class of G is 2, then $\text{cod}(G) = \{1, p, p \cdot |G/Z(G)|^{1/2}\}$.*
- (2) *If the nilpotency class of G is 3, then $\text{cod}(G) = \{1, p, p \cdot |G/G'|^{1/2}, p \cdot |G/Z(G)|^{1/2}\}$.*

PROOF. Let G be a Camina p -group. If $\text{cl}(G) = 2$, then G is a VZ p -group such that $Z(G)$ and G/G' are elementary abelian. By Theorem 1.1, $\text{cod}(G) = \{1, p, p \cdot |G/Z(G)|^{1/2}\}$. Now, suppose $\text{cl}(G) = 3$. Since G/G' is elementary abelian, $\text{cod}_{\text{lin}}(G) = \{1, p\}$. Now, we compute $\text{cod}_{\text{nl}}(G)$. Since $\text{nl}(G) = \text{Irr}(G|Z(G)) \sqcup \text{nl}(G/Z(G))$ as a disjoint union, first take $\chi \in \text{Irr}(G|Z(G))$. Since $(G, Z(G))$ is a Camina pair, $\chi(g) = 0$ for all $g \in G \setminus Z(G)$. By [18], $\chi(1) = p^{3n/2}$, and thus, $\chi = p^{3n/2}\lambda$, for some $\lambda \in \text{Irr}(Z(G))$. Then $\ker(\chi) = \ker(\lambda)$. Since $Z(G)$ is elementary abelian and $Z(G)/\ker(\lambda)$ must be cyclic, we must have $|\ker(\lambda)| = |Z(G)|/p$. Therefore,

$$\text{cod}(\chi) = \frac{|G/\ker(\chi)|}{\chi(1)} = \frac{|G|}{|\ker(\lambda)|\chi(1)} = \frac{|G| \cdot p}{|Z(G)| \cdot |G/Z(G)|^{1/2}} = p \cdot |G/Z(G)|^{1/2}.$$

Now, suppose $\nu \in \text{nl}(G/Z(G))$. Since (G, G') is a Camina pair, $(G/Z(G), G'/Z(G))$ is also a Camina pair. By [14, Corollary 5.3], $Z(G/Z(G)) = G'/Z(G) = [G/Z(G), G/Z(G)]$, that is, $G/Z(G)$ is a Camina p -group with $\text{cl}(G) = 2$. Thus, $G/Z(G)$ is a VZ p -group such that $Z(G/Z(G))$ and $((G/Z(G))/(G'/Z(G)))$ are elementary abelian. Hence, $|\ker(\nu)| = Z(G/Z(G))/p = |G'/Z(G)|/p$. Then

$$\text{cod}(\nu) = \frac{|(G/Z(G))/\ker(\nu)|}{\nu(1)} = \frac{|G/Z(G)| \cdot p}{|G'/Z(G)| \cdot \nu(1)} = \frac{p^{3n} \cdot p}{p^n \cdot p^n} = p^{n+1} = p \cdot |G/G'|^{1/2}.$$

Therefore, $\text{cod}(G) = \{1, p, p \cdot |G/G'|^{1/2}, p \cdot |G/Z(G)|^{1/2}\}$. $\square$

3. Character Codegrees Sets of p -Groups of Order $\leq p^5$, where p is an odd prime

In this section, we compute the character codegrees set for p -groups of order $\leq p^5$, where p is an odd prime. The groups of order $\leq p^5$ ($p > 2$) are listed in [9]. There are 2 isoclinic families for groups of order p^3 , 3 for groups of order p^4 , and 10 for groups of order p^5 . In all the cases, the family Φ_1 consists of all abelian groups. The structure of groups of order p^4 and p^5 when $p = 3$ is different from when p is a prime greater than 3. We theoretically prove the results for $p \geq 5$. Since the character codegrees sets of the abelian groups can be computed by Corollary 2.2, we focus on the non-abelian groups. We reference extensively the list of presentations, nilpotency classes, character degree sets, and other data available in [9], and use its group identifiers. For completeness, we compute $\text{cod}(G)$ for the groups of order 3^n (for $n = 3, 4, 5$) using MAGMA; we summarize the results in Table 4. Henceforth in this section, p denotes a prime greater than 3, unless stated otherwise.

3.1. Character codegrees sets of p -groups of order less than p^5 .

THEOREM 3.1. *Let G be a non-abelian p -group.*

- (1) *If $|G| = p^3$, then $\text{cod}(G) = \{1, p, p^2\}$.*
- (2) *Let $|G| = p^4$ and*
 - (i) *suppose $\text{cl}(G) = 2$. If $Z(G)$ is cyclic, then $\text{cod}(G) = \{1, p, p^3\}$, or $\{1, p, p^2, p^3\}$. If $Z(G)$ is non-cyclic, then $\text{cod}(G) = \{1, p, p^2\}$.*
 - (ii) *suppose $\text{cl}(G) = 3$. Then $\text{cod}(G) = \{1, p, p^2, p^3\}$.*

PROOF. Let G be a non-abelian group. If $|G| = p^3$, then G is an extraspecial group. By Corollary 2.4, $\text{cod}(G) = \{1, p, (p \cdot p^3)^{1/2}\} = \{1, p, p^2\}$. Now, suppose $|G| = p^4$. If $\text{cl}(G) = 2$, then G belongs to the isoclinic family Φ_2 , and hence, $|Z(G)| = p^2$, $|G'| = p$ and $\text{cd}(G) = \{1, p\} = \{1, |G : Z(G)|^{1/2}\}$ (see [9, Table 4.1]). Then G is a VZ-group [8, Theorem A]. If $Z(G) \cong C_p \times C_p$, by Theorem 1.1, $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \{p \cdot |G/Z(G)|^{1/2}\} = \text{cod}_{\text{lin}}(G) \cup \{p^2\}$. By Corollary 2.2, $\text{cod}_{\text{lin}}(G) = \{p^i : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$. Hence, we need $\exp(G/G')$. Since $Z(G)$ and $G/Z(G)$ are elementary abelian, $\exp(G)$ divides p^2 . Then $\exp(G/G')$ is either p or p^2 . In both the cases, we get $\text{cod}(G) = \{1, p, p^2\}$.

When $Z(G)$ is cyclic, then $G = \phi_2(31)$, or $\phi_2(211)b$. Since G is a VZ p -group, by Theorem 1.1, $\text{cod}(G) = \text{cod}_{\text{lin}}(G) \cup \{p \cdot (|G||Z(G)|^{1/2})\} = \text{cod}_{\text{lin}}(G) \cup \{p^3\}$. We compute the exponent of G/G' for each group separately. If

$$G = \phi_2(31) = \langle \alpha, \alpha_1, \alpha_2 \mid [\alpha_1, \alpha] = \alpha^{p^2} = \alpha_2, \alpha_1^p = \alpha_2^p = 1 \rangle,$$

then $G' = \langle \alpha^{p^2} \rangle \cong C_p$ and $G/G' = \langle \alpha G', \alpha_1 G' \rangle \cong C_{p^2} \times C_p$. Hence, $\exp(G/G') = p^2$ and thus, $\text{cod}(G) = \{1, p, p^2, p^3\}$. Lastly, if

$$G = \phi_2(211)b = \langle \alpha, \alpha_1, \alpha_2, \gamma \mid [\alpha_1, \alpha] = \gamma^p = \alpha_2, \alpha^p = \alpha_1^p = \alpha_2^p = 1 \rangle,$$

then $G' = \langle \gamma^p \rangle \cong C_p$ and $G/G' = \langle \alpha G', \alpha_1 G', \gamma G' \rangle \cong C_p \times C_p \times C_p$. Then $\exp(G/G') = p$, and thus, $\text{cod}(G) = \{1, p, p^3\}$.

Finally, if $\text{cl}(G) = 3$, then G is a maximal class p -group. By [5, Theorem A], $\text{cod}(G) = \{1, p, p^2, p^3\}$. This completes the proof. $\square$

From Theorem 3.1, for any non-abelian group of order p^3 , $\text{cod}(G) = \{1, p, p^2\}$. However, we obtain different possibilities for $\text{cod}(G)$ when G is a non-abelian group of order p^4 . To avoid confusion, we list down the character codegrees set corresponding to each group of order p^4 in Table 1.

Table 1: Character codegrees set of non-abelian groups of order p^4

Group G	$\text{cod}(G)$
$\phi_2(211)a$, $\phi_2(1^4)$, $\phi_2(22)$, $\phi_2(211)c$	$\{1, p, p^2\}$
$\phi_2(31)$	$\{1, p, p^2, p^3\}$
$\phi_2(211)b$	$\{1, p, p^3\}$
$\phi_3(211)a$, $\phi_3(1^4)$, $\phi_3(211)b_r$ (for $r = 1, \nu$)	$\{1, p, p^2, p^3\}$

3.2. Character codegrees sets of p -groups of order p^5 . In this subsection, we deal with the groups of order p^5 . We begin with a few results on non-abelian groups of order p^5 of nilpotency class greater than 2 in Lemma 3.2 and 3.3.

LEMMA 3.2. *Let G be a non-abelian group of order p^5 such that $\text{cl}(G) \geq 3$ and $\text{cd}(G) = \{1, p\}$.*

- (1) *If $Z(G)$ is non-cyclic, then $\text{cod}(G) = \{1, p, p^2, p^3\}$.*

(2) If $Z(G)$ is cyclic and

- (a) $\text{cl}(G) = 3$, then $|\text{cod}(G)| = 4$ and $1, p, p^4 \in \text{cod}(G)$. Further, if G/G' is not elementary abelian, then $\text{cod}(G) = \{1, p, p^2, p^4\}$.
- (b) $\text{cl}(G) = 4$, then $\text{cod}(G) = \{1, p, p^2, p^3\}$.

PROOF. Since G is nontrivial, $1, p \in \text{cod}(G)$. Also, $\text{cl}(G) \geq 3$ implies that $G \in \Phi_3 \cup \Phi_7 \cup \Phi_7 \cup \Phi_8$ (see [9]). For all such G , $\exp(G) \leq p^3$, which implies that $\exp(G/G') \leq p^3$. Thus, by Corollary 2.2, $\text{cod}(\eta) \leq p^3$, for any $\eta \in \text{lin}(G)$. Now, let χ be a nonlinear irreducible character of G . If $|G/\ker(\chi)| \leq p^2$, then $G/\ker(\chi)$ is abelian and hence $G' \subseteq \ker(\chi)$, which is absurd. Hence, $|G/\ker(\chi)| > p^2$, and thus, $|\ker(\chi)| \leq p^2$.

First, assume that $Z(G)$ is non-cyclic. Since χ cannot not be faithful, $p \leq |\ker(\chi)| \leq p^2$, which implies that $p^2 \leq \text{cod}(\chi) \leq p^3$. By [7, Theorem 1.2], $\text{cl}(G) \geq 3$ implies that $|\text{cod}(G)| > 3$. Thus, we must have $\text{cod}(G) = \{1, p, p^2, p^3\}$ (since 1 and p are already in $\text{cod}(G)$).

Now, suppose $Z(G)$ is cyclic. If $\text{cl}(G) = 4$, then G is a maximal class p -group with $\text{cd}(G) = \{1, p\}$. By [4, Theorem A], $\text{cod}(G) = \{1, p, p^2, p^3, p^4\}$. Now, suppose $\text{cl}(G) = 3$. Since $Z(G)$ is cyclic, there exists a faithful irreducible character, say δ , of G . Then $\text{cod}(\delta) = p^4$. Again, since $1, p \in \text{cod}(G)$, we must have $|\text{cod}(G)| > 3$ (from [7, Theorem 1.2]). By [4, Theorem A], $\text{cod}(G) \neq \{1, p, p^2, p^3, p^4\}$, that is, $|\text{cod}(G)| < 5$. Hence, $|\text{cod}(G)| = 4$. Further, if G/G' is not elementary abelian, then from [7, Corollary 2.5], $p^2 \in \text{cod}(G)$. Therefore, $\text{cod}(G) = \{1, p, p^2, p^4\}$. This completes the proof. $\square$

LEMMA 3.3. Let G be a non-abelian group of order p^5 such that $\text{cl}(G) \geq 3$ and $\text{cd}(G) = \{1, p, p^2\}$. Then $\text{cod}(G) = \{1, p, p^2, p^3\}$.

PROOF. Since G is nontrivial, $1, p \in \text{cod}(G)$. Proceeding along the lines of the proof of Lemma 3.2, $\text{cl}(G) \geq 3$ implies that $\exp(G/G') \leq p^3$. Thus, by Corollary 2.2, $\text{cod}(\eta) \leq p^3$, for any $\eta \in \text{lin}(G)$. Let χ be a nonlinear irreducible character of degree p in G . Then, $|G/\ker(\chi)| > p^2$, and thus, $|\ker(\chi)| \leq p^2$. If χ is not faithful, then $|\ker(\chi)| = p$, or p^2 . If $\chi(1) = p$, then $\text{cod}(\chi) = p^3$, or p^2 . Else if $\chi(1) = p^2$, then $\text{cod}(\chi) = p^2$, or p . On the other hand, suppose χ is faithful. By [19, Lemma 22], $\chi(1) = p^2$. Then $\text{cod}(\chi) = |G|/\chi(1) = p^3$. Thus, $\text{cod}(\chi) \leq p^3$ for all $\chi \in \text{nl}(G)$. Since $1, p \in \text{cod}(G)$ and $\text{cl}(G) \geq 3$, we get $|\text{cod}(G)| = 4$ by [7, Theorem 1.2]. Therefore, $\text{cod}(G) = \{1, p, p^2, p^3\}$. $\square$

REMARK 3.4. The groups of order p^5 are classified into 10 isoclinism families, Φ_i for $1 \leq i \leq 10$, where Φ_1 consists of all abelian groups (see [9]). The groups with nilpotency class 2 belong to the family Φ_i for $i = 2, 4, 5$, those of nilpotency class 3 belong to the family Φ_j for $j = 3, 6, 7, 8$, and the maximal class groups belong to the family Φ_k for $k = 9, 10$. For $G \in \Phi_i$, $i = 2, 3, 4, 6, 9$, $\text{cd}(G) = \{1, p\}$; for $G \in \Phi_5$, $\text{cd}(G) = \{1, p^2\}$, and for $G \in \Phi_j$, $j = 7, 8, 10$, $\text{cd}(G) = \{1, p, p^2\}$. Note that for $G \in \Phi_2 \cup \Phi_5$, $\text{cd}(G) = \{1, |G/Z(G)|^{1/2}\}$, that is, G is a VZ p -group. We utilize Theorem 1.1 and Corollary 2.4 to obtain $\text{cod}(G)$ when $G \in \Phi_2 \cup \Phi_5$. To deal with the groups belonging to the other families, we utilize Lemma 3.2 and 3.3.

We obtain the following results. We begin with the VZ families Φ_2 and Φ_5 in Lemma 3.5 and 3.7.

LEMMA 3.5. Let G be a group of order p^5 belonging to the family Φ_2 such that

- (i) $Z(G) \cong C_{p^3}$. Then $\text{cod}(G) = \{p^i, p^4 : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.
- (ii) $Z(G) \cong C_{p^2} \times C_p$. Then $\text{cod}(G) = \{p^i, p^2, p^3 : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$, or $\{p^i, p^3 : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.
- (iii) $Z(G) \cong C_p^3$. Then $\text{cod}(G) = \{p^i, p^2 : i = 0, 1, \dots, \log_p(\exp(G/G'))\}$.

PROOF. Let G be a group of order p^5 belonging to the family Φ_2 . Then $|Z(G)| = p^3$ and $\text{cd}(G) = \{1, p\} = \{1, |G/Z(G)|^{1/2}\}$. Hence, G is a VZ p -group where $Z(G) \cong C_{p^3}$, $C_{p^2} \times C_p$, or C_p^3 . Then, the result follows from Theorem 1.1. $\square$

REMARK 3.6. Here, we mention the information required to compute the exact $\text{cod}(G)$ when G is a group of order p^5 belonging to Φ_2 . We segregate the groups based on their center types here.

Case I - $Z(G) \cong C_{p^3}$: In this case, $G \in \{\phi_2(41), \phi_2(311)b\}$.

Case II - $Z(G) \cong C_{p^2} \times C_p$: Take

$$X_1 = \{\phi_2(221)b, \phi_2(2111)d, \phi_2(32)a_2, \phi_2(311)c\} \text{ and}$$

$$X_2 = \{\phi_2(31), \phi_2(2111)b, \phi_2(32)a_1, \phi_2(221)c\}.$$

Then $G \in X_1 \cup X_2$, when $Z(G) \cong C_{p^2} \times C_p$. For all $G \in X_1$, we can choose $a, b \in G$ such that $Z(G) = \langle a, b \rangle \cong C_{p^2} \times C_p$ and $G' = \langle b \rangle \cong C_p$, whereas, for all $G \in X_2$, we can choose $a, b \in G$ such that $Z(G) = \langle a, b \rangle \cong C_{p^2} \times C_p$ and $G' = \langle a^p \rangle \cong C_p$.

Case III - $Z(G) \cong C_p^3$: In this case, $G \in \{\phi_2(221)a, \phi_2(2111)a, \phi_2(2111)c, \phi_2(1^5), \phi_2(221)d\}$.

By Lemma 3.5, we just need the exponent of G/G' for each $G \in \Phi_2$ to compute $\text{cod}(G)$. For each group $G \in \Phi_2$, we display the corresponding $\exp(G/G')$ and $\text{cod}(G)$ in Table 2.

LEMMA 3.7. *Let G be a non-abelian group of order p^5 belonging to the family ϕ_5 . Then $\text{cod}(G) = \{1, p, p^3\}$.*

PROOF. From [9, Section 4.5], if $G \in \Phi_5$, then $G' = Z(G) \cong C_p$ and $\text{cd}(G) = \{1, p^2\}$. Thus, $\text{cd}(G) = \{1, |G/Z(G)|^{1/2}\}$, that is, G is a VZ p -group. Since $Z(G) = G'$, G is also an extraspecial p -group. By Corollary 2.4, $\text{cod}(G) = \{1, p, p^3\}$ $\square$

We present the character codegrees set for each VZ p -group of order p^5 in Table 2.

Table 2: Character codegrees set of VZ p -groups of order p^5

Group G	$\exp(G/G')$	$\text{cod}(G)$	Group G	$\exp(G/G')$	$\text{cod}(G)$
$Z(G) \cong C_{p^3}$					
$\phi_2(41)$	p^3	$\{1, p, p^2, p^3, p^4\}$	$\phi_2(311)b$	p^2	$\{1, p, p^2, p^4\}$
$Z(G) \cong C_{p^2} \times C_p$					
$\phi_2(221)b,$ $\phi_2(2111)d$	p^2	$\{1, p, p^2, p^3\}$	$\phi_2(32)a_2,$ $\phi_2(311)c$	p^3	$\{1, p, p^2, p^3\}$
$\phi_2(311)a,$ $\phi_2(32)a_1,$ $\phi_2(221)c$	p^2	$\{1, p, p^2, p^3\}$	$\phi_2(2111)b$	p	$\{1, p, p^3\}$
$Z(G) \cong C_p^3$					
$\phi_2(221)a,$ $\phi_2(2111)c,$ $\phi_2(221)d$	p^2	$\{1, p, p^2\}$	$\phi_2(2111)a,$ $\phi_2(1^5)$	p	$\{1, p, p^2\}$
$Z(G) \cong C_p$					
$\phi_5(2111),$ $\phi_5(1^5)$	p	$\{1, p, p^3\}$	-	-	-

LEMMA 3.8. *Let G be a group of order p^5 belonging to the family Φ_3 . Then $\text{cod}(G) = \{1, p, p^2, p^3\}$, or $\{1, p, p^2, p^4\}$.*

PROOF. For $G \in \Phi_3$, $\text{cl}(G) = 3$, $Z(G) \cong C_{p^2}$, or $C_p \times C_p$, and $\text{cd}(G) = \{1, p\}$. If the center is non-cyclic, then $\text{cod}(G) = \{1, p, p^2, p^3\}$ from Lemma 3.2. If the center is cyclic, then we have the following groups: $\phi_3(311)a$, $\phi_3(311)b_r$ ($r = 1, \nu$), and $\phi_3(2111)c$. It is easy to check that if $G = \phi_3(311)a$, or $\phi_3(311)b_r$ ($r = 1, \nu$), then G/G' is not elementary abelian. From Lemma 3.2, we get that $\text{cod}(G) = \{1, p, p^2, p^4\}$.

Now, we are left with

$$G = \phi_3(2111)c = \langle \alpha, \alpha_1, \alpha_2, \alpha_3, \gamma \mid [\alpha_1, \alpha] = \alpha_2, [\alpha_2, \alpha] = \gamma^p = \alpha_3, \alpha^p = \alpha_i^p = 1 \ (i = 1, 2, 3) \rangle.$$

Here $G' = \langle \alpha_2, \gamma^p \rangle \cong C_p \times C_p$, $G/G' \cong C_{p^3}$ and $Z(G) = \langle \gamma \rangle \cong C_{p^2}$. From Corollary 2.2, $\text{cod}(\eta) = p$ if $1 \neq \eta \in \text{lin}(G)$. We now compute $\text{cod}_{\text{nl}}(G)$. Let $\chi \in \text{nl}(G)$. If χ is faithful (existence of such an irreducible character is insured by the cyclic center), then $\text{cod}(\chi) = p^4$. Now, assume that χ is not faithful. Take $A = \langle \alpha_1, \alpha_2, \gamma \rangle \cong C_p \times C_p \times C_{p^2}$. Then A is a maximal abelian subgroup of G containing G' . By

[1, Theorem 2], $\chi = \lambda \uparrow_A^G$ for some $\lambda \in \text{lin}(A)$ with $\ker(\lambda) = D$ where $D \leq A$ such that A/D is cyclic. Note that $\ker(\chi) = \text{Core}_G(\ker(\lambda))$. If we take $\ker(\lambda) = D = \langle \alpha_1, \gamma \rangle$, then $\ker(\chi) = \text{Core}_G(\ker(\lambda)) = \langle \gamma \rangle$. Hence, $|\ker(\chi)| = p^2$ and thus, $\text{cod}(G) = p^2$. Now, we prove that for any $\chi \in \text{nl}(G)$, $\text{cod}(\chi) \neq p^3$, that is, $|\ker(\chi)| \neq p$. Since $\exp(A) = p^2$ and $A/D = A/\ker(\lambda)$ is cyclic, we get $|A/\ker(\lambda)| \leq p^2$. Hence, $|\ker(\lambda)| \geq p^2$. On the other hand, $|\ker(\lambda)| < p^4$, otherwise $\ker(\lambda) = A$ and hence $G' \subset \text{Core}_G(\ker(\lambda))$ which is a contradiction. Therefore, $|\ker(\lambda)| = |D| = p^2$, or p^3 . If $|D| = p^2$, then $A/D \cong C_{p^2}$. By [17, Lemma 4.2], $\langle \gamma \rangle \cap D = 1$, that is, $Z(G) \cap D = 1$. This implies that $Z(G) \cap \text{Core}_G(D) = 1$, and hence, $\text{Core}_G(D) = 1$. Hence, χ is faithful, which is a contradiction.

Now, assume that $|D| = p^3$. If $D \cong C_p^3$, then $D = \langle \alpha_1, \alpha_2, \gamma^p \rangle$, and hence $G' \subset D$. This implies that $G' \subset \text{Core}_G(D)$, and thus, D is not a suitable choice for $\ker(\lambda)$. Now, suppose $D \cong C_{p^2} \times C_p$. Then $D \in \{T_{ij}, T_k : 1 \leq i, j, k \leq p\}$, where $T_{ij} = \langle \gamma \alpha_1^i, \alpha_1^j \alpha_2 \rangle \cong C_{p^2} \times C_p$ and $T_k = \langle \gamma \alpha_2^k, \alpha_1 \rangle \cong C_{p^2} \times C_p$ for $1 \leq i, j, k \leq p$. Through routine computation, we obtain that T_{ij} is normal in G for $i = p, 1 \leq j \leq p$, and $\text{Core}_G(T_{ij}) \cong C_{p^2}$ when $1 \leq i < p, 1 \leq j \leq p$. Similarly, $\text{Core}_G(T_k) \cong C_{p^2}$ for all $1 \leq k \leq p$. Hence, when $|D| = p^3$, then D has to belong to the set $\{T_{ij}, T_k : 1 \leq i < p, 1 \leq j, k \leq p\}$, to be a candidate for $\ker(\lambda)$. Therefore, $\ker(\chi) = \text{Core}_G(\ker(\lambda)) \cong C_{p^2}$, and hence, $\text{cod}(\chi) = p^2$. Therefore, $\text{cod}(G) = \{1, p, p^2, p^4\}$. $\square$

LEMMA 3.9. *Let G be a group of order p^5 belonging to the family Φ_4 . Then $\text{cod}(G) = \{1, p, p^2\}$, $\{1, p, p^3\}$ or $\{1, p, p^2, p^3\}$.*

PROOF. For $G \in \Phi_4$, $\text{cl}(G) = 2$, $G' = Z(G) \cong C_p \times C_p$, $G/G' \cong C_p^3$, and $\text{cd}(G) = \{1, p\}$. From Corollary 2.2, $\text{cod}(\eta) = p$ if $1 \neq \eta \in \text{lin}(G)$. We now compute $\text{cod}_{\text{nl}}(G)$ for each group $G \in \Phi_4$. We deal with each group in the family separately. We start with

$$G = \phi_4(221)a = \langle \alpha, \alpha_1, \alpha_2, \beta_1, \beta_2 \mid [\alpha_1, \alpha] = \beta_1 = \alpha_1^p, [\alpha_2, \alpha] = \beta_2 = \alpha^p, \alpha_2^p = \beta_1^p = \beta_2^p = 1 \rangle.$$

Here $G' = Z(G) = \langle \alpha^p, \alpha_1^p \rangle \cong C_p \times C_p$. Take $A = \langle \alpha^p, \alpha_1, \alpha_2 \rangle \cong C_p \times C_{p^2} \times C_p$. Then A is a maximal abelian subgroup of G containing G' . Let $\chi \in \text{nl}(G)$. By [1, Theorem 2], $\chi = \lambda \uparrow_A^G$ for some $\lambda \in \text{lin}(A)$ with $\ker(\lambda) = D$ where $D \leq A$ such that A/D is cyclic. It is easy to see that, just as in the case of $\phi_3(2111)c$ in the proof of Lemma 3.8, $|\ker(\lambda)| = |D| = p^2$, or p^3 . If $D = \langle \alpha_1, \alpha_2 \rangle$, then A/D is cyclic. In this case, $\ker(\chi) = \text{Core}_G(\ker(\lambda)) = \text{Core}_G(D) = \langle \alpha_1 \rangle$. Hence, $|\ker(\chi)| = p^2$ which implies that $\text{cod}(\chi) = p^2$. Now, if $D = \langle \alpha^p, \alpha_2 \rangle$, then A/D is cyclic. Here, $\ker(\chi) = \text{Core}_G(\ker(\lambda)) = \langle \alpha^p \rangle$. Hence, $|\ker(\chi)| = p$ which implies that $\text{cod}(\chi) = p^3$. Thus, $\text{cod}_{\text{nl}}(G) = \{p^2, p^3\}$, and finally, $\text{cod}(G) = \{1, p, p^2, p^3\}$.

Using similar arguments, we conclude that $\text{cod}(G) = \{1, p, p^2, p^3\}$ if $G \in \{\phi_4(221)b, \phi_4(221)d_r \ (r = 1, 2, \dots, (p-1)/2), \phi_4(221)e, \phi_4(2111)b, \phi_4(2111)c\}$.

Now, suppose

$$G = \phi_4(221)c = \langle \alpha, \alpha_1, \alpha_2, \beta_1, \beta_2 \mid [\alpha_1, \alpha] = \beta_1 = \alpha_1^p, [\alpha_2, \alpha] = \beta_2 = \alpha_2^p, \alpha^p = \beta_1^p = \beta_2^p = 1 \rangle.$$

Here $G' = Z(G) = \langle \alpha_1^p, \alpha_2^p \rangle \cong C_p \times C_p$. Take $B = \langle \alpha_1, \alpha_2 \rangle \cong C_{p^2} \times C_{p^2}$. Then B is a maximal abelian subgroup of G containing G' . Let $\nu \in \text{nl}(G)$. Again, by [1, Theorem 2], $\nu = \theta \uparrow_B^G$ for some $\theta \in \text{lin}(B)$ with $\ker(\theta) = E$ where $E \leq B$ such that B/E is cyclic. If $E = \langle \alpha_1 \rangle$, then B/E is cyclic. Further, $\ker(\nu) = \text{Core}_G(E) = E$ which implies that $|\ker(\nu)| = p^2$. Hence, $\text{cod}(\nu) = p^2$. Now, we prove that $\text{cod}_{\text{nl}}(G) = \{p^2\}$. Just as in the case of $\phi_3(2111)c$ in the proof of Lemma 3.8, $p^2 \leq |E| \leq p^3$. However, if $|E| = p^3$, then owing to the fact that $\exp(A) = p^2$, we get that $E \cong C_{p^2} \times C_p$. Then E contains all the p -ordered elements of A , and hence, $G' = Z(G) \subset E$. Thus, G' is contained in $\text{Core}_G(E)$ as well, and hence, E cannot be a candidate for $\ker(\theta)$. Therefore, $|E| = p^2$. Clearly, $E \not\cong C_p \times C_p$ (otherwise, $G' = E$). Thus, $E \cong C_{p^2}$. By [21, Theorem 4.2], total number of possibilities for such a subgroup E is $p^2 + p$. In fact, $E \in \{\langle \alpha_1 \alpha_2^i \rangle, \langle \alpha_1^{pj} \alpha_2 \rangle \mid 1 \leq i \leq p^2, 1 \leq j \leq p\}$. It is routine exercise to check that in all the cases, $\text{Core}_G(E) = E$, that is, E is normal in G (due to the fact that $[\alpha_1, \alpha] = \alpha_1^p$ and $[\alpha_2, \alpha] = \alpha_2^p$). Therefore, $|\ker(\nu)| = p^2$. Hence, $\text{cod}_{\text{nl}}(G) = \{p^2\}$, and finally, $\text{cod}(G) = \{1, p, p^2\}$.

Using similar arguments, we conclude that $\text{cod}(G) = \{1, p, p^2\}$ if $G = \phi_4(2111)a$, or $\phi_4(1^5)$.

Now, let

$$G = \phi_4(221)f_0 = \langle \alpha, \alpha_1, \alpha_2, \beta_1, \beta_2 \mid [\alpha_1, \alpha] = \beta_1, [\alpha_2, \alpha] = \beta_2 = \alpha_1^p, \alpha_2^p = \beta_1^p, \alpha^p = \beta_1^p = \beta_2^p = 1 \rangle.$$

Here $G' = Z(G) = \langle \alpha_1^p, \alpha_2^p \rangle \cong C_p \times C_p$. Take $T = \langle \alpha_1, \alpha_2 \rangle \cong C_{p^2} \times C_{p^2}$. Then T is a maximal abelian subgroup of G containing G' . Let $\phi \in \text{nl}(G)$. Again, by [1, Theorem 2], $\phi = \psi \uparrow_T^G$ for some $\psi \in \text{lin}(T)$ with $\ker(\psi) = J$ where $J \leq T$ such that T/J is cyclic. If $J = \langle \alpha_1 \rangle$, then T/J is cyclic. Further, $\ker(\phi) = \text{Core}_G(J) = \langle \alpha_1^p \rangle \cong C_p$ which implies that $|\ker(\phi)| = p$. Hence, $\text{cod}(\phi) = p^3$. Now, we prove that $\text{cod}_{\text{nl}}(G) = \{p^3\}$. Now, same as in the case of $\phi_4(221)c$, J must be isomorphic to C_{p^2} . Then J belongs to the set $\{\langle \alpha_1 \alpha_2^i \rangle, \langle \alpha_1^{pj} \alpha_2 \rangle \mid 1 \leq i \leq p^2, 1 \leq j \leq p\}$. Since $[\alpha_1, \alpha] = \alpha_1^p$ and $[\alpha_2, \alpha] = \alpha_2^p$, it is easy to check that $\text{Core}_G(E) \cong C_p$ in all the cases. That is, $|\ker(\phi)| = p$, which implies that $\text{cod}(\phi) = p^3$. Therefore, $\text{cod}_{\text{nl}}(G) = \{p^3\}$, and finally, $\text{cod}(G) = \{1, p, p^3\}$. Using similar arguments, we conclude that $\text{cod}(G) = \{1, p, p^3\}$ if $G = \phi_4(221)f_r$ for $r = 1, 2, \dots, (p-1)/2$.

Hence, $\text{cod}(G) = \{1, p, p^2\}$, $\{1, p, p^3\}$ or $\{1, p, p^2, p^3\}$ if $G \in \Phi_4$ with $|G| = p^5$. $\square$

LEMMA 3.10. *Let G be a group of order p^5 belonging to the family Φ_i for $i = 6, 7, 8, 10$. Then $\text{cod}(G) = \{1, p, p^2, p^3\}$.*

PROOF. From [9, Section 4.5], if $G \in \Phi_6$, then $\text{cl}(G) \geq 3$, $\text{cd}(G) = \{1, p\}$ and $Z(G) \cong C_p \times C_p$. By Lemma 3.2, $\text{cod}(G) = \{1, p, p^2, p^3\}$. Finally if $G \in \Phi_6 \cup \Phi_8 \cup \Phi_{10}$, then $\text{cl}(G) \geq 3$ with $\text{cd}(G) = \{1, p, p^2\}$. Then $\text{cod}(G) = \{1, p, p^2, p^3\}$ by Lemma 3.3. $\square$

LEMMA 3.11. *Let G be a group of order p^5 belonging to the family Φ_9 . Then $\text{cod}(G) = \{1, p, p^2, p^3, p^4\}$.*

PROOF. From [9, Section 4.5], if $G \in \Phi_9$, then $\text{cl}(G) = 4$ and $\text{cd}(G) = \{1, p\}$. Then $\text{cod}(G) = \{1, p, p^2, p^3, p^4\}$ by [4, Theorem A]. $\square$

Table 3: Character codegrees set of non-VZ p -groups of order p^5

Group G	$\text{cod}(G)$	Group G	$\text{cod}(G)$
$\phi_3(2111)a$, $\phi_3(2111)b_r$ ($r = 1, \nu$), $\phi_3(1^5)$, $\phi_3(221)a$, $\phi_3(221)b_r$ ($r = 1, \nu$), $\phi_3(2111)d$, $\phi_3(2111)e$	$\{1, p, p^2, p^3\}$	$\phi_3(311)a$, $\phi_3(311)b_r$ ($r = 1, \nu$), $\phi_3(2111)c$	$\{1, p, p^2, p^4\}$
$\phi_4(221)a$, $\phi_4(221)b$, $\phi_4(221)d_r$ ($r = 1, 2, \dots, (p-1)/2$), $\phi_4(221)e$, $\phi_4(2111)b$, $\phi_4(2111)c$, $\phi_4(2111)e$	$\{1, p, p^2, p^3\}$	$\phi_4(221)c$, $\phi_4(2111)a$, $\phi_4(1^5)$	$\{1, p, p^2\}$
$\phi_4(221)f_0$, $\phi_4(221)f_r$ ($r = 1, 2, \dots, (p-1)/2$)	$\{1, p, p^3\}$	-	-
All Groups in $\Phi_6 \cup \Phi_7 \cup \Phi_8 \cup \Phi_{10}$	$\{1, p, p^2, p^3\}$	-	-
All Groups in Φ_9	$\{1, p, p^2, p^3, p^4\}$	-	-

3.3. Verification of results through MAGMA. We have constructed a MAGMA function `CharacterCodegree` to verify $\text{cod}(G)$ for a given group G . With its help, we verify the data recorded in Table 1, 2 and 3 for primes 5, 7, 11 and 13. The code is publicly available in MAGMA via a GitHub repository [23].

3.4. Character codegree sets for groups of order 3^n where $n = 3, 4, 5$. We utilize the MAGMA code `CharacterCodegree` to obtain the character codegrees of non-abelian groups of order 3^n (for $n = 3, 4, 5$); we present the values in Table 4. The group identifiers are with respect to the SMALLGROUPS library [2].

Table 4: The set $\text{cod}(G)$ for the non-abelian groups of order 3^n (for $n = 3, 4, 5$)

$\text{cod}(G)$	SmallGroup ID
$\{1, 3, 9\}$	[27, 3], [27, 4]
$\{1, 3, 9\}$	[81, 3], [81, 4], [81, 12], [81, 13]
$\{1, 3, 27\}$	[81, 14]
$\{1, 3, 9, 27\}$	[81, 6], [81, 7], [81, 8], [81, 9], [81, 10]
$\{1, 3, 9\}$	[243, 2], [243, 32], [243, 33], [243, 37], [243, 38], [243, 47], [243, 62], [243, 63]
$\{1, 3, 27\}$	[243, 44], [243, 45], [243, 64], [243, 65], [243, 66]
$\{1, 3, 9, 27\}$	[243, 3], [243, 4], [243, 5], [243, 6], [243, 7], [243, 8], [243, 9], [243, 11] [243, 12], [243, 13], [243, 14], [243, 15], [243, 17], [243, 18], [243, 21], [243, 22] [243, 28], [243, 29], [243, 30], [243, 34], [243, 35], [243, 36], [243, 39], [243, 40] [243, 41], [243, 42], [243, 43], [243, 46], [243, 49], [243, 51], [243, 52], [243, 53] [243, 54], [243, 56], [243, 57], [243, 58], [243, 59], [243, 60], [243, 62], [243, 63]
$\{1, 3, 9, 81\}$	[243, 16], [243, 19], [243, 20], [243, 50], [243, 55]
$\{1, 3, 9, 27, 81\}$	[243, 24], [243, 25], [243, 26], [243, 27]

We summarize the results obtained in Section 3 in Theorem 3.12.

THEOREM 3.12. *Let G be a non-abelian p -group, where p is an odd prime.*

- (1) *If $|G| = p^3$, then $\text{cod}(G) = \{1, p, p^2\}$.*
- (2) *If $|G| = p^4$, then $\text{cod}(G) \in \{\{1, p, p^2\}, \{1, p, p^3\}, \{1, p, p^2, p^3\}\}$.*
- (3) *If $|G| = p^5$, then $\text{cod}(G) \in \{\{1, p, p^2\}, \{1, p, p^3\}, \{1, p, p^2, p^3\}, \{1, p, p^2, p^4\}, \{1, p, p^2, p^3, p^4\}\}$.*

Acknowledgements

The author thanks Sunil Kumar Prajapati for carefully reading the many versions of this paper and elaborate discussions on the content. The author also acknowledges the University of Auckland for providing remote access to their computational facilities.

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY KATTANKULATHUR, CHENNAI-603203, TAMIL NADU, INDIA
Email address: udeepayush@gmail.com